

APSMO 2010 Set 4

1 Problems

Problem 1.1 Allie has half as much money as Ben. Ben has \$3 more than Emma. Emma has 5 times as much money as Shauna. Shauna has \$1.
How much money do Allie and Ben have together?

Problem 1.2 Following only the paths shown, what is the number of different paths that go from A to B to C to D and touch each of those points exactly once?



Problem 1.3 Sarah and Tyler ride their bikes. They start at the same time from the same point and ride in the same direction. Sarah travels 20km every hour, and Tyler travels 15km every hour.
At the end of how many hours will Sarah be 30km ahead of Tyler?

Problem 1.4 Michael has some cards. If he puts them in 5 equal piles, there are 3 left over. If he puts them in 4 equal piles, there are 2 left over. If he puts them in 3 equal piles, there is 1 card left over.
What is the fewest cards Michael may have?

Problem 1.5 The figure shown is made up of 6 congruent squares. The perimeter of the figure is 42cm. It is folded along the dotted lines to form a box.
How many 1cm cubes can fit in the box?

